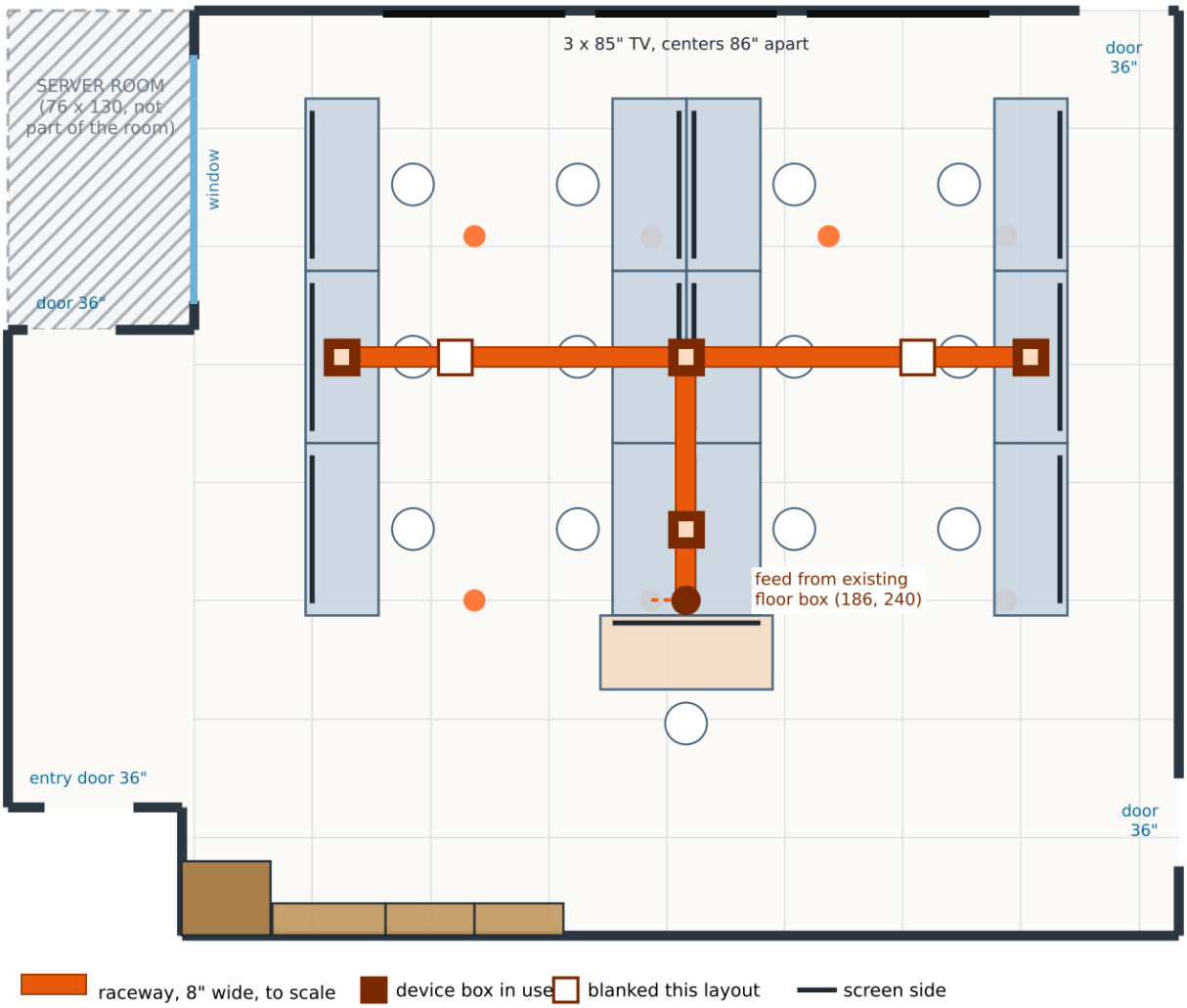
Room 413: FSR Smart-Way Floor Raceway Concept

One set of runs that serves all three layouts. Concept for discussion with the district electrician and facilities, not an engineered drawing.

Layout 1: Class

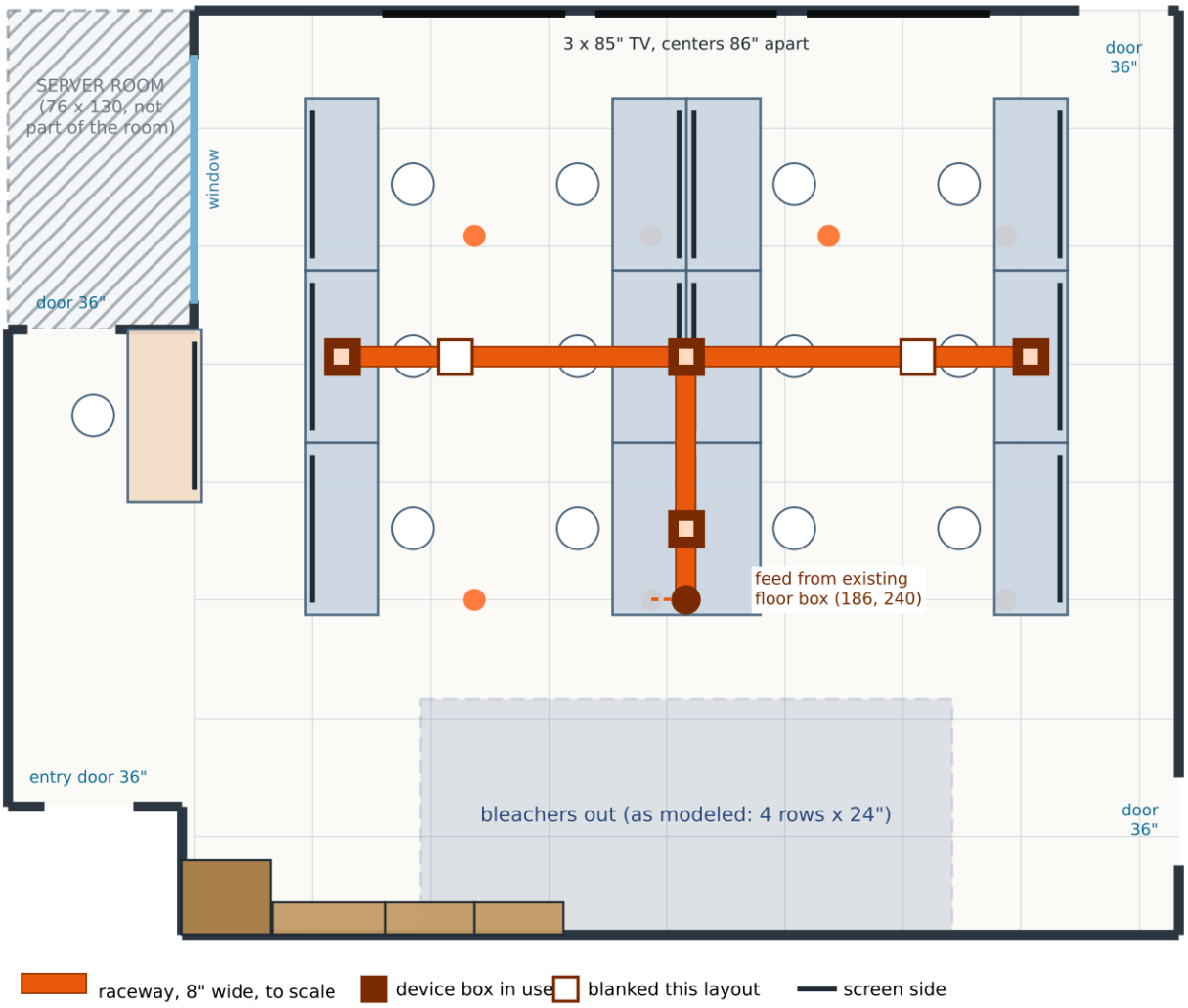
Bleachers folded. Teacher desk T'd onto the south end of the center stack.



The idea. A single 280 in. cross-run along z = 141 in. passes through the middle desk of each group in the class layout (west column, center stack, east column) and through the middle desk of each team row in the competition layout, which happen to land at the same height in the room. A 99 in. spur runs south from the center of the cross-run to the south end of the stack and picks up the existing floor boxes as the feed. That puts a device box within one desk-length (70 in.) of every seat in every layout.

Layout 2: Class + bleachers out

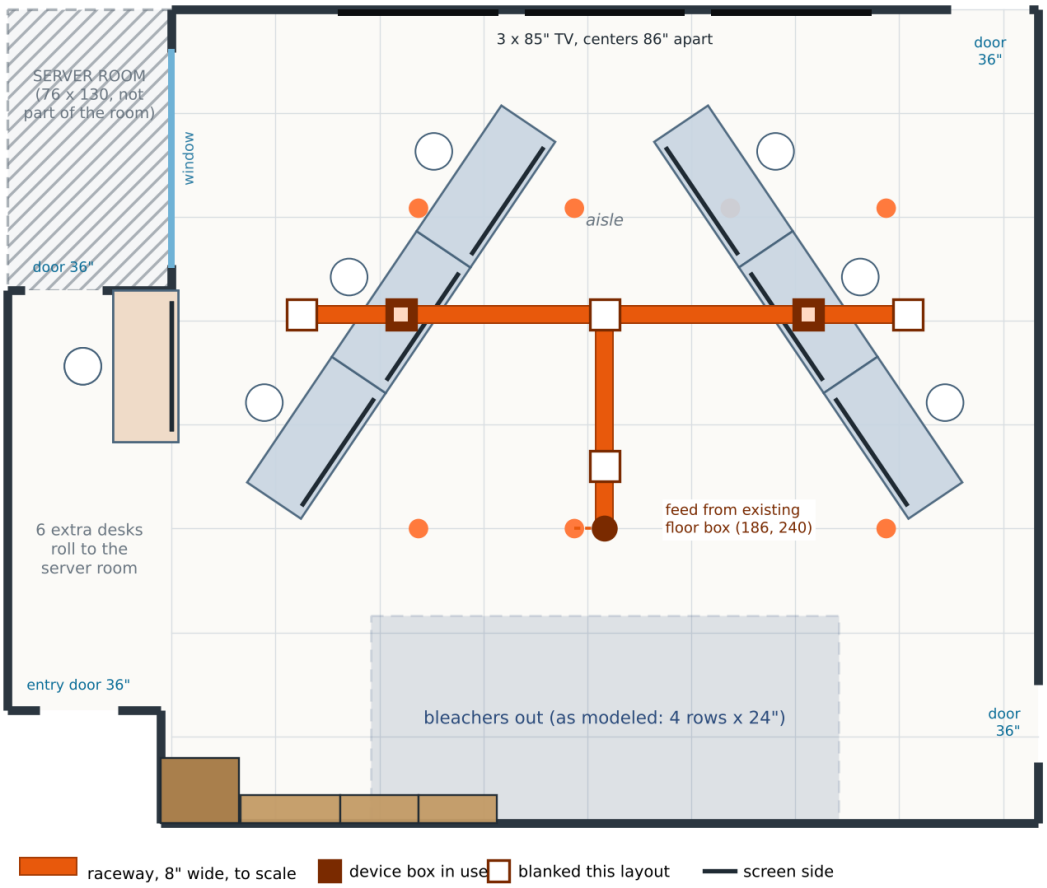
Teacher desk moves to the alcove; the student layout is unchanged.



The student layout is the same as Layout 1, so the same boxes stay live. The teacher desk moves to the alcove, and the bleacher footprint (shown as modeled, 4 rows at 24 in.) stays clear of the raceway, which ends at z = 240 in.

Layout 3: Competition

Two 1x3 team rows at 56 degrees, 46" aisle; extra desks leave the room.



Only the two boxes under the middle desk of each team row are live. The other positions are blanked, and the spur sits in the center of the 46 in. aisle, where it is walked on but not rolled across by desks.

Rough quantities and open items

Item	Estimate
Raceway, cross-run + spur	about 380 in. (31.6 ft), roughly six 6 ft sections cut to length
Fittings	1 tee (at the center of the cross-run), end caps, feeder tie-in at the existing floor box (186, 240)
Device boxes	6 positions: 4 live in the class layouts, 2 live in competition; the rest blanked
Open: height and width	FSR's page does not list the raceway cross-section. The 8 in. width drawn here is the figure we recalled for the product; confirm before final layout.
Open: rolling loads	Straight sections are rated 8,000 lb per tire; the tee is rated 1,500 lb and corners 2,500 lb. Keep caster traffic off fittings where possible.
Open: electrical	Circuit count, breaker capacity and code review for esports PCs on the run. UL listing was not stated on the product page.
Open: feed and network	The feed point is assumed to be the existing floor box. Network can share the low-voltage compartment but needs a patch path from the server room.

Source: FSR Smart-Way Floor Raceway. Positions are in inches from the northwest corner of the main room (x east, z south).